

Smart Devices for Smart Cities in India

Exploring Security Vulnerabilities of E-Waste Management

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Abstract—Smart city is an emerging concept in India where Government of India have started initiatives towards building smart cities through new projects. Given the technological advancement in the conceptualization of smart cities, e-waste management has become extremely crucial construct that needs attention at this moment in the country, specifically for the cities to qualify as a smart city. Smart devices such as smart phones, tablets, smart-toys, wireless chargers, and others are to name a few which are very common in urban life and are building blocks of smart cities. However, such devices have end-of-life (EoL) which build the e-waste after exhaustion of its lifespan and thus their systematic disposal becomes a necessity. For our study, we analyzed and evaluated three smart devices (smart phones, tablets, and smart-toys) and explored their potential security threats through user focused interviews. We present two case studies featuring potential threats from e-waste recycling hotspot and mobile repair cafe. Unfortunately, e-waste management is often neglected and our study shows that the lack of preventive measures lead to serious data security threats as well. Such security vulnerabilities may lead to the collapse of smart cities which lead to monetary losses. Through our research, we propose an upgraded Life-Cycle Framework focusing on Indian second-hand device market to ensure systematic e-waste management.

Index Terms—Smart Devices, Smart City, E-waste Management, Security Threats, Supply Chain, Indian Second-Hand Device Market.

I. INTRODUCTION

Urban India is developing rapidly and smart devices are becoming an integral part of this development [40]. History is the testament of the urban settlements, which show such technologically advanced establishment have been key activity centres of business, wealth, idea, art, and entrepreneurship [42]. India is not different from any other country that realized the rapid urbanization in the past is crucial for its economic development. One of the most important and noteworthy steps taken by the Government of India (GoI) towards technological advancement is the *smart city* mission [9]. As described by the ISO, there are specific requirements for a city to be termed as a smart city, such as energy efficiency (ISO 37100); water quality and waste water treatment (ISO 24510); road traffic

safety (ISO 39001); social security (ISO 22313); data security (ISO/IEC 27551) etc. [28]. Very recently, ISO have published the standard on indicators for smart cities which provides the cities a set of indicators which allows to measure their performance [36]. It is important to note that the role of ISO is important in developing, designing and monitoring performance of a Smart city.

Smart devices such as smart phones, tablets, wireless charger, and others are elemental constructive and operative units of smart cities. With the increasing environmental concerns, one of the major requirements for smart city establishments is sustainable management of electronic waste [13]. Based on the smart devices, cloud networks, and emerging internet of things; the network topology and communication mediums is in a dynamic shift [43]. Conversely, with more technologically advanced devices emerging and impacting our lives positively, we must be aware of the their End of life (EoL) which contribute a majority of the e-waste [18, 19, 20]. Due to lack of knowledge and poor management of such e-waste potential security threats and vulnerabilities are emerging especially due to the esoteric and unstable nature of the EoL of such devices [18]. Security threats can arise from several factors, including but not limited to, improper disposal of e-waste, unsupervised data destruction of storage devices, use of insecure public cloud services, and unaware end-users [21]. Potential security vulnerabilities can lead to personal data leakage such as login credentials [15], contact list [14], messages [23], photos [38], and other crucial information of the owner and their people in their contact list which can often be obtained from discarded personal devices such as mobile phones [22, 44].

Previous studies have shown several reverse engineering techniques which can be deployed to exploit several electronic devices to obtained personal information [39]. Roychowdhury et al. discussed the threats that can arise from e-waste as well as the electronic components. Their work emphasizes on reverse engineering approaches of threats and proposed anti-reverse engineering solutions [41]. The uniqueness of this

study is the identification of the electronic component mixing problem in the supply chain of electronic items which is critical to mitigate such risks from the deployment instance [41]. Recently, focused work on device specific security threat has been attempted by the authors. Mobile phones have been chosen for this purpose. The authors identified crucial paths in the supply chain of waste mobile phones which were not mentioned in earlier studies. The identified paths in the supply chain are critical as security threat may arise from those [21]. While there is a flooding increase in security research among the data scientists, it has been rare occurrence where smart devices have been targeted for evaluation for security threats. The research questions that arise are - What are possible security threats arising from smart devices in a smart city? What are the different issues and challenges associated with those devices? What is the role of the second hand market in generating security threat from these devices? In this study, we have explored the existing security vulnerabilities of three smart devices: smart phones, tablets, and smart toys by interviewing users of these devices in the Indian second-hand device marketplace. Issues and challenges has been identified and discussed with respect to the aforementioned devices with the smart city context in India, specifically focusing on the second hand-market and e-waste management. An sustainable life cycle based framework has been proposed as a solution to mitigate the risks.

II. SMART CITY IN INDIA

Govt. of India has taken smart city mission to develop 100 cities across the country [34]. The term became a buzz word in India soon after Prime Minister of India declared about the smart city mission [31]. The main objectives of this program are city improvement, redevelopment of city extension and green field development. It is mainly citizen friendly and sustainable [31]. For the first time this program has been taken on the basis of competition method through select cities for funding and using as area based development. This spirit will lead to competitive and co-operative feudalism [1, 31]. This mission expects active participation of smart people in governance and reform program. Smart people must involve themselves in execution of smart solution through implementation of reforming programs doing more and designing post project structure. One of the approach taken for development of smart city and ensure public participation is by introducing Special Purpose Vehicle via increased use of ICT and mobile based tools [1]. Schemes and initiatives under Digital India mission such as Aadhaar, digital lockers and computerization of land records are gaining popularity by playing a constant role in smart city mission [2]. The following are some smart solutions has been proposed under smart city initiatives. Some of those are - Bhubaneswar E-Mobility Plan; Promoting Innovative Smart Solutions under Smart Cities Mission, AMRUT and Swachh Bharat Mission; Soil-Bio-Technology (SBT) based Sewage Treatment Plants etc [3].

III. STUDY METHODOLOGY

The study is focused on security threats from three smart devices, namely - smart phone, tablet and smart toys. The study identifies the potential threat in these smart toys which are often used in smart cities. End-of-life (EoL) smart devices may be vulnerable to certain threats. Detailed literature review was carried out understand the current situation as well the potential threats. The literature were skimmed to identify potential threats. Two field visits were carried out with semi-structured questionnaire in informal e-waste hotspot and mobile repairing centres.

The mobile repair centres were found to be ignorant in answering the questions but some information was gathered with their minimum responses. They were asked questions regarding their processes, whether they have access to customers data, what is their model of business etc. Based on the outcome a supply chain based model was developed which depicts the current situation of EoL smart device management.

IV. SECURITY THREATS FROM SMART DEVICES

Security threats for smart devices can either be caused by attackers and hackers or threats might arise from the unawareness of the user [29]. End of life smart devices are also vulnerable to such threats. Earlier studies have shown potential security threat from electronics devices and e-waste [8, 41]. This section discusses about security threats from selected smart devices.

A. Security Threats from Smart Phones

The smart phones that we use today are very powerful and employ wireless multimedia communications [46]. These smart phones are susceptible to vulnerabilities and security threats [35]. A smart phone may have several vulnerabilities such as implementation errors, incompatibility, unawareness of user, vulnerabilities from wireless network and other external objects etc [29]. These vulnerabilities and the threats give rise to attacks in a smart phone [33, 10]. Threats can be caused by attackers in different ways, such as malware, wireless network attack, denial of service and break in [29]. Other types of threats are generally caused by the lack of awareness from the user end or sometimes it is caused intentionally [33, 29]. Attacks can happen due to loss of mobile phone, platform alteration, and sometimes due to malfunctioning of the device [26]. Smart phones are also vulnerable to phishing attack [17, 16] which is often done by creating a spoof version of a particular web page [24, 44]. This is a good example where the security threat is both due to unawareness of user as well as intentional attack. Man-in-the-middle attacks are also a good probability when user connects to internet using Wi-Fi hotspots [44].

B. Security Threats from Tablets

Tablets or Tab, a popular smart device, have similar security threats with the smart phones. Considering the user platform to be Android platform, which is widely used across the globe, the following security threats can arise.

- Tablets are susceptible to wireless network attacks. This can be achieved by any attacker via modification of information on the wireless network by sniffing, spoofing or eavesdropping [29, 33].
- A hacker can get full or partial control of a tablet using flaw of code, code injection etc [29].
- Malware can expose private information from the storage device of a tablet. Additionally, they can induce unwanted and meaningless code execution; and abuse of costly services and applications [29].
- Tablets are also prone to Phishing attacks as described in the previous subsection. These spoofing spoofing websites tricks the users by manipulating the display constraints in the smart devices. The browser hides the URL bar and uses the complete screen of the device to display the spoof website which looks like the original one. This makes such attacks quite powerful [24, 44].

C. Security Threats from Smart Toys

Smart Toys are electronic devices that are classified as smart as they are able to perform some operations more efficiently compared to general toys [11]. Inclusion of complex electronic devices, state-of-the-art features and quirky marketing strategies have made these smart toys successful [32]. Majority of the smart toys are Wi-Fi enabled and can connect to the internet [30]. The systems designed for smart toys are not cross-checked with standard security protocols designated for electronic devices [45]. Due to this, the toys are exposed to a huge security threat. The attacker can be a man-in-the-middle who can remotely capture the network traffic between a device and the cloud. The attacker is also able to record and replay network packets [47]. A study by Ng et al. focuses on two security issues in smart toys - a) location history and data tracking and b) encryption and data security [37]. The nature of the toy computing and threat architecture affects privacy as it can reveal the identity of a child and location of the child. Additionally, sensitive information related to the child or the child's family can be at stake along networking capabilities of the device through which collected data could be shared [12]. According to de Carvalho et al. spoofing, tampering, repudiation, information disclosure, denial of service and elevation of privilege are the potential threats arising from smart toys [12].

V. CASE STUDIES

In this section two case studies has been presented - a) E-waste hotspot in Kolkata and b) Unauthorized mobile repair cafe which shows current situation and the possible security threats.

A. Case Study A: E-waste Hotspot in Kolkata

This is a local hotspot in Kolkata, West Bengal situated near the Anwar Shah Road area. There are more than 5 shops in this area who buy electronic waste along with other dry waste from the local waste scavengers known as Kabadiwalas. They are quite expert in identifying good electronics from others. Due to this reasons, they are sometimes called the



Fig. 1. Unauthorized Mobile Repair Centre

cherry pickers. These people collect electronic waste from households and several other places in exchange of certain amount of money. The amount varies based on the product quality. Generally, small and medium sized e-waste is collected and sold to these shops. Each of these shops has at least one person who dismantles e-waste to recover the printed circuit board (PCB). Mobile phones and other smart devices are not dismantled and sold as it is to some unauthorized mobile repairers who try to make the devices functional by using loose electronic components and other parts recovered from other such devices. These PCBs are sold either per kg or per piece basis (based on the source and weight of the PCB) to some dealers whose identity was not revealed. Basically, these unidentified persons are middle men who collect PCBs on behalf of the informal recyclers.

B. Case Study B: Unauthorized mobile repair cafe

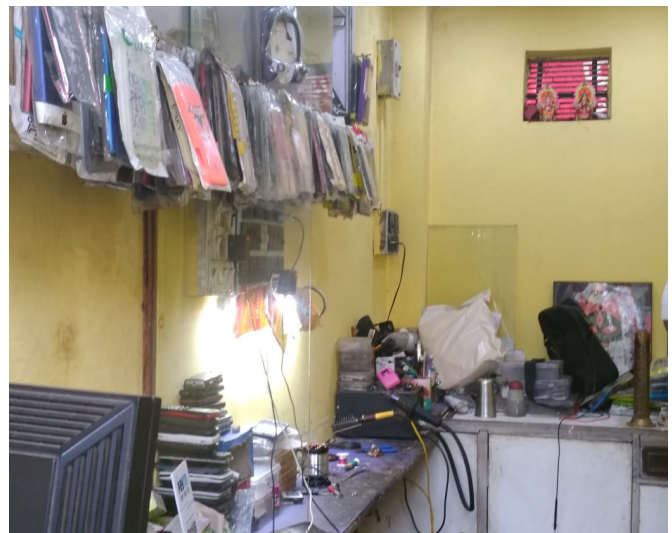


Fig. 2. Unauthorized Mobile Repair Centre

This is a unauthorized mobile repair shop situated in Kolkata, West Bengal in the Tollygunje area. There are near by three more shops in this area who repair mobile phones without any license. These people have basic training on electronic repairing . They used instruments like hot air gun, forceps, soldering iron etc for repairing purpose. These people can repair mobile phones at a very cheap rate compared to authorized mobile phone service centres. Eventually most customers are attracted towards such repairing centres. They provide their services on pay-per-visit scheme and does not provide any guarantee of their services. Sometimes they replace some non-functional parts from the devices and they dispose them with Municipal Solid Waste. If any printed circuit board is there with them, they sell it to some buyers whose identity was not disclosed.

VI. LIFE CYCLE BASED MODEL FOR SMART DEVICES IN INDIA

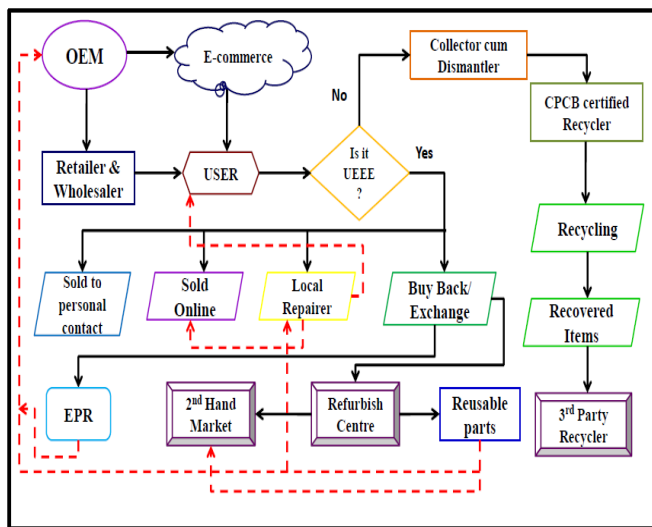


Fig. 3. Life cycle based model for Smart Devices in India

A model has been developed following the previous work [21] where a life cycle based framework was provided for small WEEE. This model has been extended for the smart devices mentioned earlier. Figure 3 represents the proposed model for smart devices. The current model is also a life cycle based model which considers most of the stakeholders along the supply chain. Original equipment manufacturers (OEMs), retailers, wholesalers, e-commerce sectors, collector cum dismantlers, certified recyclers and second hand market has been considered as important actors for the overall life cycle of smart devices and included in the model. The user procures the smart devices either from e-commerce sites or from shops. When the device reaches its end of life, the question is whether it is repairable or not. If it is repairable then it can be routed to the following paths a) sold to a close relative or friend; b) sold through online websites; c) repairing by local unauthorized repairer; d) disposal through buy-back or exchange program and e) refurbishing and reselling in the second hand market. Other option include repairing and

reusing by the user, repairing and then selling it online and sending back to the OEM through EPR program. If the device is beyond repairing, it is generally collected by the collector cum dismantler, who route it to the recyclers for recycling, material recovery and resource circulation. Recovered material streams are handled by respective third party recyclers.

VII. FINDINGS AND DISCUSSIONS

The research carried out in terms of literature review and interviews reveals that there exist a strong supply chain that operates behind the vibrant second hand market. The general mass is unaware of the security threats that can arise from the EoL smart devices. The life cycle based model clearly explains the path and back-loops of material flow which are most often untraceable. This is one of the important findings as these are the cases from where the security threat arise. In general, due to poverty and time management issues, the devices are routed towards the local shops for repairing. But the price of privacy and security which is compromised is too much. It is evident that the practice of selling the used electronics either to relative or via websites such as OLX and eBay is vibrant in India. This is more than a century old practice in India even before internet was born. The smart city mission in India have a lot of indicators, based on which it is decided whether it qualify as smart city or not [4].

Considering the ISO guidelines for Sustainable city which is nothing but a global version of smart city, hazardous waste management is a very important indicator [27]. In view of that, e-waste management is important and the EoL smart devices are basically e-waste. Currently, there are 178 e-waste recyclers are there who have been certified by the Central Pollution Control Board (CPCB) but a very few is operating at their full capacity [5, 25]. The new e-waste rules 2016, which is primarily based on EPR has been published and amended in 2018 by the Ministry of Environment, Forest and Climate Change (MoEFCC) however the implementation and monitoring of it is questionable [6, 7, 25]. As the model suggests, the check point of whether the device is e-waste or not is the most crucial point which decides whether it will reach the desired facility or not. In that way, the situation is not so bright and needs more intervention to address the issues related to security of these smart devices. The threats not only exist after EoL but also during its functional period.

All these needs to be addressed via proper amendment of the e-waste rule and awareness campaigns. The responsibility of the OEMs must be revised so as to facilitate better implementation of EPR and Producer Responsibility for smart devices. There should be some basic precautions to be taken by the users and a general guideline needs to be published by the Government of India or by the Bureau of Indian Standards. Last but not least, the whole power lies in the hands of the users and they need to be more careful and aware of the issues and device strategies accordingly. Industry-academia collaboration is a necessity as it can further enhance the securities of the smart devices and thereby making them less

vulnerable towards any attack. Additionally, more research work is required in the field of science-policy framework.

VIII. CONCLUSION

In this study, security threats of three smart devices which are commonly used in smart cities (smart phones, tablets and smart toys) has been evaluated. The work is based on literature support, interview outcomes and case studies. Potential threats evolving from these selected smart devices has been identified and the type of attacks that can happen has also been summarized. The case studies provides the current situation of smart device repairing and disposal. The extended life cycle based model for smart devices in the smart city depicts the paths and back-loops of material flow. This is a prima facie for further model development for EoL electronics in a smart city. Solutions to tackle the security issues has been discussed on the backdrop of Indian scenario of smart cities. This study has focused on a less explored research area of security threat from EoL electronics, smart devices specifically and certainly more research will enrich this area. The findings are expected to be helpful for policymakers and researchers.

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